

Day 5 — Loop Practice in Python

Detailed Seminar Notes | Tables • Sums • Patterns • Syntax • Dry Runs • Rules • Practice Programs

Day 5 focuses on **practising loops**. After learning **for loop** and **while loop** in Day 4, the next step is to use loops to solve real problems like printing multiplication tables, finding sums, generating patterns, and writing number-based programs.

1) Goal of Day 5

- Strengthen understanding of **for loop** and **while loop**.
- Use loops to solve **repetition-based problems**.
- Learn how to trace loop variables step by step.
- Build confidence for seminar explanation and coding practice.

2) Quick revision of loop syntax

for loop syntax

```
for variable in sequence:  
    statement(s)
```

while loop syntax

```
while condition:  
    statement(s)  
    update
```

Remember: **indentation is compulsory**, and in a **while loop** the variable must be updated so that the condition eventually becomes False.

3) Loop Practice Topic 1 — Printing numbers

The simplest loop practice starts with printing numbers in different orders.

Program 1: Print numbers from 1 to 10

```
for i in range(1, 11):  
    print(i)
```

Output: 1 2 3 4 5 6 7 8 9 10

Program 2: Print numbers from 10 to 1

```
for i in range(10, 0, -1):  
    print(i)
```

Here the step value is **-1**, so the numbers decrease.

Program 3: Print even numbers from 2 to 20

```
for i in range(2, 21, 2):  
    print(i)
```

Program 4: Print odd numbers from 1 to 19

```
for i in range(1, 20, 2):  
    print(i)
```

4) Loop Practice Topic 2 — Multiplication tables

Tables are one of the most common beginner loop programs.

Program 5: Multiplication table of 5

```
n = 5
for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

Output:

```
5 x 1 = 5
5 x 2 = 10
...
5 x 10 = 50
```

Program 6: Multiplication table of user input

```
n = int(input("Enter a number: "))
for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

This is a better seminar example because it shows **input + loop + multiplication** together.

Dry Run for table of 3

Iteration	i	Expression	Printed line
1	1	3 * 1	3 x 1 = 3
2	2	3 * 2	3 x 2 = 6
3	3	3 * 3	3 x 3 = 9
...	...	...	...
10	10	3 * 10	3 x 10 = 30

5) Loop Practice Topic 3 — Sum of numbers

Loops are very useful for adding numbers one by one.

Program 7: Sum of first 5 natural numbers

```
sum = 0
for i in range(1, 6):
    sum = sum + i
print("Sum =", sum)
```

Output: Sum = 15

How it works

Iteration	i	Old sum	New sum = sum + i
1	1	0	1
2	2	1	3
3	3	3	6
4	4	6	10
5	5	10	15

Program 8: Sum of numbers from 1 to n

```
n = int(input("Enter n: "))
sum = 0
for i in range(1, n + 1):
```

```

    sum = sum + i
    print("Sum =", sum)

```

If the user enters **10**, the program adds 1+2+3+...+10 and prints 55.

6) Loop Practice Topic 4 — Count and total separately

Sometimes we use one loop to calculate **sum**, and another variable to calculate **count**.

Program 9: Count even numbers from 1 to 20

```

count = 0
for i in range(1, 21):
    if i % 2 == 0:
        count += 1
print("Count =", count)

```

This prints **10** because there are 10 even numbers from 1 to 20.

Program 10: Sum of even numbers from 1 to 20

```

sum = 0
for i in range(1, 21):
    if i % 2 == 0:
        sum += i
print("Sum =", sum)

```

7) Loop Practice Topic 5 — Factorial

Factorial of a number **n** means multiplication of all numbers from 1 to n.

Example: $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

Program 11: Factorial of a number

```

n = int(input("Enter a number: "))
fact = 1
for i in range(1, n + 1):
    fact = fact * i
print("Factorial =", fact)

```

Dry Run for n = 4

Iteration	i	Old fact	New fact = fact * i
1	1	1	1
2	2	1	2
3	3	2	6
4	4	6	24

8) Loop Practice Topic 6 — Pattern programs

Pattern programs use loops to print shapes using stars or numbers. They are very common in exams and seminars.

Program 12: Print 5 stars in one line

```

for i in range(5):
    print("*", end=" ")

```

Program 13: Print stars in 5 lines

```

for i in range(5):

```

```
print("")
```

Program 14: Right triangle star pattern

```
for i in range(1, 6):  
    print(" " * i)
```

Output:

```
*  
**  
***  
****  
*****
```

Explanation of pattern logic

- In the first iteration, **i = 1**, so `" " * 1` prints one star.
- In the second iteration, **i = 2**, so two stars are printed.
- This continues until **i = 5**.

Program 15: Reverse star pattern

```
for i in range(5, 0, -1):  
    print(" " * i)
```

Output:

```
*****  
****  
***  
**  
*
```

9) Using while loop for practice programs

The same practice problems can also be solved using **while loop**.

Program 16: Print 1 to 5 using while

```
i = 1
while i <= 5:
    print(i)
    i += 1
```

Program 17: Sum of first 5 numbers using while

```
i = 1
sum = 0
while i <= 5:
    sum += i
    i += 1
print("Sum =", sum)
```

Program 18: Table of 4 using while

```
n = 4
i = 1
while i <= 10:
    print(n, "x", i, "=", n * i)
    i += 1
```

10) Important rules while solving loop problems

- Decide **what repeats** in the problem.
- Choose **for loop** if the number of repetitions is known.
- Choose **while loop** if repetition depends on a condition.
- For sum problems, start with **sum = 0**.
- For multiplication / factorial problems, start with **fact = 1** or **product = 1**.
- For counting problems, start with **count = 0**.
- Update the loop variable properly in a while loop.
- Trace the program with a small value first to check logic.

11) Common beginner mistakes in loop practice

- Using the wrong **range()** limits, like **range(1, 10)** when 10 should also be included.
- Starting factorial with **0** instead of **1**.
- Forgetting to reset **sum**, **count**, or **fact** before the loop.
- Using **print()** inside the loop when the final answer should be printed after the loop.
- Not updating **i** in while loop.
- Mixing tabs/spaces or wrong indentation.

12) Difference between common loop variables

Variable	Used for	Initial value	Example
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i	Loop counter / iteration	Usually 0 or 1	for i in range(1, 6)
sum	Adding values	0	sum = sum + i
count	Counting items	0	count += 1
fact	Factorial / multiplication	1	fact = fact * i

13) Practice questions for Day 5

- Print numbers from 1 to 50.
- Print numbers from 50 to 1.
- Print all even numbers between 1 and 100.
- Print all odd numbers between 1 and 50.
- Find the sum of first 10 natural numbers.
- Find the sum of even numbers from 1 to 20.
- Find the factorial of 6.
- Print multiplication table of 7.
- Print a right triangle pattern of 6 rows.
- Write all the above using **while loop** also.

14) Viva / seminar questions

- Why do we use loops in practice programs?
- How is sum logic different from factorial logic?
- Why does factorial start with 1 and not 0?
- How do we print a multiplication table using a loop?
- What is the use of **end=' '** in print()?
- What is the difference between counting and summing?
- How are pattern programs connected to loops?

15) Short seminar summary

Day 5 is about using loops to solve actual coding problems. We practiced **printing numbers, multiplication tables, sums, factorials, and patterns**. The main idea is to identify what repeats, choose the correct loop, initialize helper variables properly, and trace the program carefully.

16) Quick revision box

Topic	Key point
Printing numbers	Use range() with correct start, stop, and step
Table program	Loop from 1 to 10 and multiply by the number
Sum program	Initialize sum = 0 and keep adding
Factorial	Initialize fact = 1 and keep multiplying
Patterns	Use loop count to decide number of stars / numbers

while loop practice	Update loop variable every iteration
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